Agent Generated Hypothesis and Research Plan

Novel Hypothesis:

Metformin inhibits glutaminase activity in GBM cells, leading to decreased glutamine utilization and impaired cell growth and survival. This effect may be enhanced by the hypoxic microenvironment of GBM, which increases the reliance on glutamine metabolism. Furthermore, this glutamine metabolism inhibition by metformin could sensitize GBM cells to temozolomide or other chemotherapeutic agents.

Research Plan:

This research plan will test the hypothesis and incorporate both *in vitro* and *in vivo* experiments, along with mechanistic explorations.

- *In vitro* studies:
- Cell lines: Use several GBM cell lines (e.g., U87-MG, U251-MG, LN229) and a non-tumorigenic astrocyte cell line as a control.
- Treatment: Treat cells with varying concentrations of metformin (e.g., 0, 1, 5, 10 mM) under both normoxic (21% O2) and hypoxic (1% O2) conditions.
- Assays:
- Cell viability assay (e.g., MTT or CellTiter-Glo) to measure cell growth inhibition.
- Glutaminase activity assay to measure the effect of metformin on glutaminase activity.
- Glutamine uptake assay to measure glutamine utilization.
- Western blot to measure protein expression levels of glutaminase (GLS), AMPK, and downstream targets involved in cell survival and apoptosis.
- Measurement of intracellular glutamine and glutamate levels using LC-MS.
- Combination studies with temozolomide (TMZ) to assess synergistic effects. Measure cell viability after treatment with metformin, TMZ, and the combination. Calculate combination index (CI) values. CI < 1 indicates synergism.
- *In vivo* studies:
- Animal model: Use orthotopic xenograft model by injecting GBM cells (e.g., U87-MG) into the brains of immunocompromised mice (e.g., NSG mice).
- Treatment: Treat mice with metformin (e.g., via oral gavage) or vehicle control. A group treated with temozolomide and a group treated with the combination of metformin and temozolomide should also be included.
- Monitoring: Monitor tumor growth using MRI or bioluminescence imaging.
- Endpoint: Measure survival time.
- Tissue analysis: After euthanasia, collect tumor tissue for immunohistochemistry to assess glutaminase expression, cell proliferation (Ki-67), apoptosis (cleaved caspase-3), and AMPK activation. Perform metabolomics analysis on tumor tissue to quantify glutamine and glutamate levels.
- Mechanism Exploration

- Assess the impact of metformin treatment on the expression of genes involved in glutamine metabolism using RT-qPCR.
- Examine the effect of metformin on mitochondrial function, given the interplay between glutamine metabolism and the TCA cycle.
- Investigate the role of autophagy in metformin-induced cell death in GBM cells.

Novelty Assessment:

Based on the PubMed searches performed, the hypothesis appears reasonably novel. While the search for "metformin glioblastoma glutaminase" yielded no results, indicating a lack of direct research specifically linking metformin to glutaminase in GBM, the search "metformin glutamine metabolism" returned a paper (PMID:30646605) showing that metformin impairs glutamine metabolism and autophagy in tumor cells. This supports the general concept but doesn't fully explore the nuances proposed here, particularly the impact of hypoxia and synergistic effects with temozolomide in the GBM context. Therefore, while the general area is not entirely unexplored, the specific hypothesis concerning glutaminase inhibition, hypoxic enhancement, and temozolomide sensitization in GBM warrants further investigation.